

of data storage layer later gave birth to the dynamic web applications we're all so fond of.

Diffie-Hellman Key Exchange Algorithm

When the era of the internet, email and online messaging began, preventing sensitive information from going astray increasingly became important. With this context, the invention of public key cryptography was a pivotal one to ensure the safety, security and integrity of any information transmitted digitally. For making this possible, we have a lot to thank Whitfield Diffie and Martin Hellman – two pioneers of public key cryptography – for.

They devised an ingeniously simple method in the form of an algorithm to ensure a message or conversation between two parties remains confidential. This algorithm uses arithmetic modules as the basis of its calculation. In this method, two parties that have no prior knowledge of each other collectively establish a shared secret key over an insecure channel of communication. They can then use this key to encrypt the communication henceforth using a symmetric key algorithm (type of algorithm which uses the same cryptographic keys for both, encryption and decryption). The beauty of this technique is that the shared secret key is not protected from interception but it is protected from modification. This algorithm is extremely simple when it comes to its concept but has a very high level of cryptographic stability which is actually due to its mathematical complexity.

